Doolittle

$$A = \begin{pmatrix} b_1 & a_1 & & & \\ a_1 & b_2 & a_2 & & 0 \\ & a_2 & \ddots & \ddots & \\ & & \ddots & b_{n-1} & a_{n-1} \\ 0 & & & a_{n-1} & b_n \end{pmatrix} =$$

$$= \underbrace{\begin{pmatrix} 1 & & & & \\ l_1 & 1 & & & 0 \\ & l_2 & \ddots & & \\ & & \ddots & \ddots & \\ 0 & & & l_n & 1 \end{pmatrix}}_{L} \times \underbrace{\begin{pmatrix} v_1 & a_1 & & & 0 \\ & v_2 & a_2 & & \\ & & \ddots & \ddots & \\ & 0 & & \ddots & a_{n-1} \\ & & & & v_n \end{pmatrix}}_{U}$$

$b_1 = v_1 \Rightarrow v_1 = b_1$

$a_k = l_k v_k \Rightarrow l_k = a_k / v_k \quad , k = \overline{1, n-1}$

$b_k = l_{k-1} a_{k-1} + v_k \Rightarrow v_k = b_k - l_{k-1} a_{k-1}, \quad k = \overline{2, n}$

Cholesky

$$A = \begin{pmatrix} b_1 & a_1 & & & \\ a_1 & b_2 & a_2 & & 0 \\ & a_2 & \ddots & \ddots & \\ & & \ddots & b_{n-1} & a_{n-1} \\ 0 & & & a_{n-1} & b_n \end{pmatrix} =$$

$$= \underbrace{\begin{pmatrix} v_1 & & & \\ l_1 & v_2 & & 0 \\ & l_2 & \ddots & \\ & & \ddots & v_n \\ 0 & & l_{n-1} & \end{pmatrix}}_{L} \times \underbrace{\begin{pmatrix} v_1 & l_1 & & \\ & v_2 & l_2 & 0 \\ & & \ddots & \ddots \\ & & & v_n \\ 0 & & & \end{pmatrix}}_{L^T}$$

$v_1^2 = b_1 \Rightarrow$ alegem soluția pozitivă $v_1 = \sqrt{b_1}$

$v_k = \sqrt{b_k - l_{k-1}^2}$ (alegem sol poz.) , $k = \overline{2,n}$

$a_k = l_k \cdot v_k \Rightarrow l_k = \frac{a_k}{v_k}$, $k = \overline{1,n-1}$

Jacobi tridiag

$$A = \begin{bmatrix} \alpha_1 & a_1 & & & \\ c_1 & \alpha_2 & a_2 & & 0 \\ & c_2 & \ddots & \ddots & \\ & & \ddots & \ddots & a_{n-1} \\ 0 & & & c_{n-1} & \alpha_n \end{bmatrix}$$

$$x_i^{(p+1)} = \frac{b_i - \sum_{j=1, j\neq i}^{n} a_{ij} x_j^{(p)}}{a_{ii}} =$$

$$= \frac{b_i - c_{i-1} \cdot x_{i-1} - a_i \cdot x_{i-1}}{\alpha_{ii}}$$

Givens - tridiag

$$A = \begin{bmatrix} \alpha_1 & a_1 & & & \\ c_1 & \alpha_2 & a_2 & & 0 \\ & c_2 & \ddots & \ddots & \\ & & \ddots & \ddots & a_{n-1} \\ 0 & & & c_{n-1} & \alpha_n \end{bmatrix}$$